

10.3 - Compound Interest

Problem Solving, Math 1000

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1 Definition

Under simple interest, the interest rate is applied only to the principal. However, under [compound interest](#), the interest rate is applied to both the principal *and* the previously accrued interest.

All things being equal, money invested under compound interest grows much more quickly, and money lent under compound interest can be a much greater burden for a borrower to undertake.

2 Theorem (Annual Compounding Formula)

The future value, F , of P dollars *compounded annually* for t years with APR r is given by:

$$F = P(1 + r)^t$$

2.1 Example

A [certificate of deposit](#), or [CD](#), is a investment vehicle that lies somewhere between an ordinary bond and a savings account. Generally, CDs are considered very safe. Like a bond, you are essentially lending your money to the issuer of the CD for some time period. But, unlike a bond, your interest is compounded (usually on an annual basis, but other compounding periods are possible). Generally, your money is pretty well 'locked up' for the duration of the CD – withdrawing early normally will cost a reasonably steep early withdrawal penalty. At the end of a CD, you generally have either the option to cash out or roll over your balance into a new CD (with new terms and a new APR, often). The cash value of a CD at the end of its term is often referred to as its [future value](#).

Suppose you invest \$1,500 on a CD with an APR of 4.8% compounded annually. Let t denote the length (in years) of the CD's term. We will use the annual compounding formula with $P = \$1500$ and $r = 0.048$ to find the future value, F , of the CD for a few common values of t .

The formula gives:

$$F = \$1500(1 + 0.048)^t$$

Which, for the following values of t , yields:

$t = 1$	$F = \$1572.00$
$t = 2$	$F = \$1647.46$
$t = 5$	$F = \$1896.26$
$t = 10$	$F = \$2397.20$
$t = 15$	$F = \$3030.47$

It is interesting to note that the 15 year CD essentially doubles the initial investment. This brings us to a nice idea:

The Rule of 72:

There's a rough rule of thumb, called the [rule of 72](#) which estimates how long it will take an annually compounded amount to double. Given a certain APR under annual compounding, divide the number 72 by the APR in percent form (i.e. divide 72 by 12 for a 12% interest rate) to yield approximately the number of years it will take any amount invested at that rate to double.

3 Theorem (General Compounding Formula)

We can also talk about interest compounded more frequently than once per year. Given a periodic interest rate p compounded a total of T times, the future value, F , of P dollars is given by the formula

$$F = P(1 + p)^T$$

If this formula looks similar to the previous formula, that's because it is. Essentially, the only difference is that we've replaced the APR, r , with the periodic interest rate, p . We calculate p by taking the APR and dividing it by the number of compounding periods in a year. Similarly, we calculate T by multiplying the number of compounding periods per year by the total number of years, t , for which the investment is made.

All together, this amounts to another application of one of our most common themes in this course: Changing the yardstick we use to measure things in order to previous solutions work for new problems.

4 Theorem (Continuous Compounding Formula)

We can also talk about compounding interest *continually*. That is, interest compounded infinitely over infinitely small time periods. This involves perhaps an unexpected number – the irrational number $e = 2.71828...$ To understand why this number shows up, we would need some calculus. But rest assured, it does indeed work, and the formula is relatively easy to use as well.

The future value of P dollars compounded continuously for t years with APR r is given by:

$$F = Pe^{rt}$$

Because e is an irrational number (that is, it has an infinite, nonrepeating decimal expansion), we can only ever get an approximately precise numerical value of F with a calculator. However, we can make this approximately precise answer arbitrarily close to the real answer (that is, within whatever error tolerance we desire), so it's not particularly a problem.

We can use the continuous compounding formula even when the term r is a fractional part of a year. For example, for a term of 5 years and 3 months, we can use $t = 5.25$. For a term of 500 days, we can use $t = \frac{500}{365}$, and so on.

4.1 Example

Consider a CD with principal $P = \$1500$ and APR of 4.8% as before, but let us consider this time the case when the interest compounds continuously.

We will use the continuously compounding formula with $P = \$1500$ and $r = 0.048$ to find the future value, F , of the CD for a few common values of t .

The formula gives:

$$F = \$1500e^{(0.048)t}$$

Which, for the following values of t , yields:

$t = 1$	$F = \$1573.76$
$t = 2$	$F = \$1651.14$
$t = 5$	$F = \$1906.87$
$t = 10$	$F = \$2424.11$
$t = 15$	$F = \$3081.65$

This shows that continuously compounding interest isn't actually all that much better than annually compounding interest. For example, a CD with a 4.8% APR compounded quarterly is much better than a CD with a 4.7% APR compounded continuously.

5 Definition

The **annual percentage yield**, or **APY**, is the annual percentage increase on the value of an investment when both the APR and the frequency of compounding are taken into account. When comparing investments with different APRs and different compounding frequencies, the APY is the right tool to use. Normally, the simplest way to compute the APY is to find the value of \$1 at the end of one year under each choice.

Table 10-7 shows the computation of APYs for a 6% APR under various compounding frequencies.

Compounding	T	F	APY
annually	1	1.06	6.000%
semi-annually	2	$\left(1 + \frac{0.06}{2}\right)^2 = 1.0609$	6.090%
quarterly	4	$\left(1 + \frac{0.06}{4}\right)^4 \approx 1.061364$	6.1364%
monthly	12	$\left(1 + \frac{0.06}{12}\right)^{12} \approx 1.061678$	6.1678%
continuously	∞	$e^{0.06} \approx 1.061837$	6.1837%

■ **TABLE 10-7** Computation of APYs for a 6% APR